# Ayush Kumar Joshi

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## CAREER OBJECTIVE

Computer Science undergraduate (B.E. CSE, batch 2028) with a strong C++ and data structures & algorithms foundation (235+ LeetCode problems) and project experience building backend APIs in Python (Flask, FastAPI) on top of SQL databases. Built a low-light face detection system (NOVA-Face) and a full-stack climate analytics platform. Seeking a Software Development Intern role to write reliable, secure, scalable software.

## TECHNICAL SKILLS

**Languages:** Python, C++, C, Java (coursework)

**Databases:** SQL (MySQL, LiveSQL), NoSQL (MongoDB)

**Web / API:** HTML, CSS, FastAPI, Flask, REST API design, OpenCV

**Practices:** Debugging & troubleshooting, algorithm design & complexity analysis, technical documentation, Git

## SOFT SKILLS

**Communication:** Presentation skills, technical documentation, article/report writing

**Collaboration:** Teamwork, cross-functional coordination (2-person research team on NOVA-Face)

**Problem-Solving:** Debugging & troubleshooting, systematic issue diagnosis

## PROJECTS

**NOVA-Face — Night-Optimized Vision Architecture for Face Detection** *Jun 2026*

- Built a low-light face detection pipeline in Python using YOLOv8 for detection and FastAPI to serve the model, addressing the gap between models trained on normal-light images and real nighttime footage.
- Owned the backend, API, and front-end integration on a 2-person team, working from a SQL-based dataset pipeline through to a served, testable model.
- Wrote the technical documentation and literature review, turning research notes into a structured writeup.

**Climate Change Analytics & Visualization Platform — B.E. Project, Chandigarh University** *2026*

- Built a full-stack analytics platform with a React front-end and a Flask backend, using a Random Forest model for climate trend prediction and a Groq-hosted LLaMA model to answer natural-language questions about the results.
- Found and fixed an exposed API key and incorrect environment-variable handling during a pre-submission code review.
- Prepared the technical deep-dive and Q&A for the project viva and wrote the final project report.

**Image-to-Sketch Converter (Live)** *Dec 2025*

- Built a full-stack web app that converts images into pencil-style sketches, using OpenCV for image processing and a Flask API on the backend.
- Handled the image processing logic, error cases, and API design.

**IoT-Based Radar System** *Jan – Apr 2025*

- Built a real-time radar system with an HC-SR04 ultrasonic sensor and an Arduino Uno, displaying live object distance and direction on a Processing IDE interface.
- Debugged sensor noise and timing issues to get a stable, real-time data stream.

## EDUCATION

**Bachelor of Computer Science Engineering:** Chandigarh University, 2024 – 2028, CGPA: 7.93

**Senior Secondary (12th):** Green Valley School for Children, 2024

## CERTIFICATIONS

- Completed Summer Training course as part of the AI for Future Workforce – Intel digital readlines (2026)
- Introduction to Machine Learning - NPTEL (2026)
- Program Programming in Python - Coursera (2025)

## ACHIEVEMENTS

- Solved 235+ Questions on LeetCode.